⚡ DEVPOST AI BUILDERS HACKATHON 2026

Kites

Spatial In-Browser Manga & Comic Translator

100% Client-Side WebGPU Neural Pipeline • Zero Python • Zero Cloud Lock-in • Authentic Comic Lettering in Real-Time

Creator: Unheat (dangtruongan)
Live Demo: https://kites-translator.pages.dev
Source Code: https://github.com/Unheat/Kites



01 · PROBLEM STATEMENT

Reading Raw Comics is Broken and Painful

Existing manga translation tools force users into trade-offs between impossible setups and expensive paywalls.



Complex Python / CUDA Setups

Legacy desktop tools like Cotrans demand gigabytes of PyTorch weights, manual CUDA installation, and complex terminal commands that 99% of readers cannot set up.



Aggressive Cloud Paywalls

Commercial browser extensions charge $10/month recurring subscriptions, impose strict daily page caps, or sell token credits that exhaust after two manga chapters.



Severe Privacy & Security Risks

Cloud translation extensions upload user browsing habits, private bookmarks, and raw image scans to opaque remote servers without user consent.



Ugly White-Box Erasure

Crude rectangular erasers paste solid white boxes over dialogue balloons, obliterating underlying screentones, line art, and comic textures.

02 · SOLUTION OVERVIEW

Kites: 100% Local-First In-Browser AI Translation

A zero-install Chrome extension running computer vision, generative inpainting, and typography directly on your graphics card.

ZERO SETUP

Instant Browser Extension

Install in 30 seconds. Runs natively inside Google Chrome and Chromium browsers with zero terminal commands or external software.

WEBGPU AI

Client-Side Acceleration

PaddleOCR detection and LaMa neural inpainting execute directly on your device GPU via ONNX Runtime Web with zero server cost.

100% PRIVATE

Zero Data Uploads

Images are processed in local browser memory and stored in local IndexedDB. Your reading history and images never touch our servers.

COMIC AWARE

Studio Typography

Measures speech bubbles, balances multi-line wrapping, auto-sizes fonts, and renders authentic comic lettering directly in-page.

03 · TARGET USERS

Who Needs Kites: Global Manga & Web Comic Ecosystem

Serving international comic readers, social media art fans, and scanlation creators worldwide.

01

Global Manga & Webtoon Fans

CHALLENGE:

70M+ readers globally wait weeks or months for fan scanlation groups to translate raw Japanese, Korean, and Chinese releases.

HOW KITES EMPOWERS THEM:

Translate raw chapters instantly on manga sites with one click, preserving original artwork and dialogue emotion.

02

Social Media Art Fans

CHALLENGE:

Millions of short comics and illustrations are shared daily on X (Twitter), Pixiv, and Reddit by creators without official translations.

HOW KITES EMPOWERS THEM:

The floating hover button translates comic strips directly inside the social media timeline with zero screenshots or copy-pasting.

03

Scanlation Teams & Creators

CHALLENGE:

Manual cleaning, cloning, text redrawing, and typesetting in Photoshop takes 4–8 hours per manga chapter.

HOW KITES EMPOWERS THEM:

Kites Studio provides an interactive 8-point bounding box editor to polish translations, adjust fonts, and export 1:1 lossless PNGs.

04 · PRODUCT FEATURES

Frictionless In-Page Interaction & Dedicated Studio

Four versatile browsing trigger modes paired with an interactive full-screen post-editor.



In-Page Hover Translation

Hover over any comic image to reveal the floating translate button. Replaces text in-place with zero page layout distortion.

Persistent Pins & Auto-Translate

Keep action badges pinned across all comic panels, or enable Viewport Auto-Translate to queue chapters automatically as you scroll.

Built-in Kites Studio

Triple-view inspection (Original, Clean Inpaint, Typeset), 8-point interactive bounding box resizing, and lossless 1:1 PNG export.

SPA Reversion Shield

MutationObserver guards prevent modern React/SPA frameworks (X/Twitter, Reddit) from reverting translated images back to raw scans.

05 · VISUAL PROOF

Uncompromising Quality: Zero White Boxes

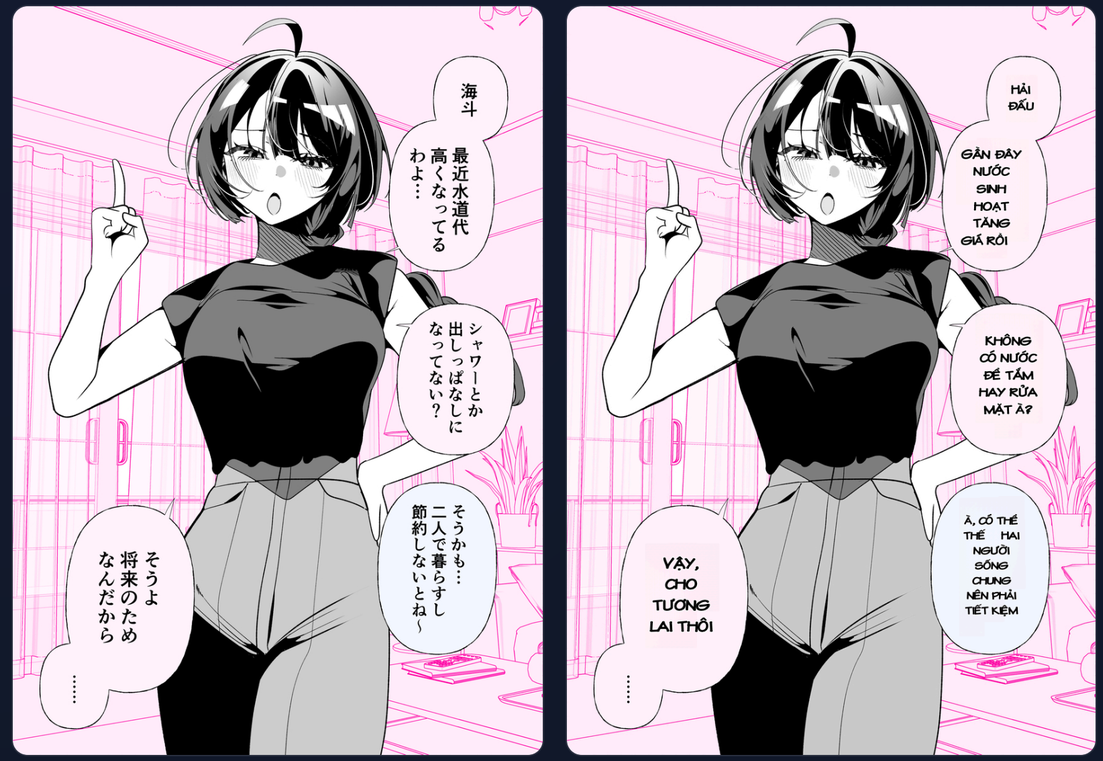
Neural background inpainting erases strictly within character contours, preserving line art and screentones.

JAPANESE ➔ ENGLISH (SATO-SAN BENCHMARK)



Original Scan (Left) vs. Kites Translated (Right)

JAPANESE ➔ VIETNAMESE (PIXIV #145272482)



Original Scan (Left) vs. Kites Translated (Right)

06 · TECHNICAL ARCHITECTURE

Manifest V3 Offscreen Document Sandboxing

Bypassing service worker lifecycle limits to run persistent WebGPU neural computation in Chromium.

1. In-Page Scanner

Content Script

- Discovers DOM images ≥150px
- Attaches CSS Anchor hover button
- Wipes responsive srcset candidates
- SPA Reversion Shield (MutationObserver)

2. Message Hub

Service Worker

- Deduplicates redundant triggers
- Enforces 1–5 task concurrency
- Token-bucket rate limiter
- Routes explicit target/source RPCs

3. WebGPU Sandbox

Offscreen Document

- Persistent context (no timeouts)
- PaddleOCR DBNet rotated detection
- Promise.all: Inpaint + Translation
- 4-Pass binary-search typesetting

4. Storage & Display

Dexie IndexedDB

- Stores clean plate + TextBlocks
- Automatic 7-day TTL cache pruning
- Converts Base64 to safe Blob URLs
- Studio editor interactive re-rendering

07 · AI TECHNOLOGIES

The 4 Core Pillars of the Kites AI Engine

Orchestrating computer vision, graph clustering, generative diffusion, and LLM inference.

COMPUTER VISION (OCR)

PaddleOCR DBNet + SVTR

Runs ONNX Runtime Web via WebGPU/WASM. Detects arbitrary rotated 4-point convex hull polygons (dt_polys) with projective perspective crop homography.

SPATIAL GEOMETRY

Kruskal Minimum Spanning Tree

Direction detection (connected-component majority scoring) + 14-stage noise battery + Kruskal MST bubble clustering to reconstruct natural reading order.

GENERATIVE INPAINTING

LaMa Manga & AOT-GAN

Zero-resizing 1:1 scale dynamic 64px patch bucketing. Erases strictly along character contour polygons with UNCLIP 1.8 expansion to preserve line art.

LANGUAGE TRANSLATION

WebLLM & Cloudflare Shared Pool

Runs on-device WebLLM (Qwen 2.5, Llama 3.2) or zero-cost Cloudflare Worker pool with Google OAuth and multi-provider fallback (Mistral -> Gemma -> Groq).

08 · IMPACT & VALUE

Democratizing Visual Storytelling at Zero Server Cost

A sustainable, highly scalable consumer product engineered for privacy and accessibility.

$0

Marginal Server Cost

Compute offloaded to client-side WebGPU

100%

User Privacy Guaranteed

Zero images uploaded to remote servers

< 3s

Average Pipeline Latency

Parallel inpainting + translation execution

WITHOUT KITES (STATUS QUO)

- $10/month cloud subscription fees with harsh daily token limits
- Painful terminal setup (Python, PyTorch, CUDA, Git dependencies)
- Crude white boxes destroying comic backgrounds and line art
- Personal reading history uploaded to unknown third-party servers

WITH KITES (LOCAL-FIRST ERA)

- 100% Free & Unlimited reading with zero monthly subscriptions
- One-click Chrome extension install (no terminal or coding required)
- Clean LaMa neural inpainting strictly preserving original art
- 100% On-device privacy with zero cloud image uploads

09 · ROADMAP & FUTURE VISION

Scaling Kites: Next Milestones & Community Launch

Expanding platform reach, multi-language OCR coverage, and creator tooling.

Q3 2026

COMPLETED

Core Engine & v1.0.3

- WebGPU ONNX PaddleOCR & LaMa Inpainting
- Kruskal-MST spatial bubble clustering
- Interactive Kites Studio 8-point editor
- Pre-built zip & Chrome Web Store submission

Q4 2026

IN PROGRESS

Store Launch & Browsers

- Chrome Web Store official public release
- Firefox MV3 & Microsoft Edge packaging
- Dynamic WebLLM model download manager
- Custom comic typography font packs

Q1 2027

PLANNED

Asian OCR Expansion

- Vertical Korean Hangul dedicated dictionary
- Traditional Chinese Manhua tuning
- Canvas / WebGL viewport screen snipping
- Community scanlation cloud sync

Try Kites: https://kites-translator.pages.dev • **GitHub:** https://github.com/Unheat/Kites • **Contact:** andangtruong085@gmail.com